Leakage-Aware Deep Learning for Financial Event-State Forecasting

Jose Ignacio Valdivia Aguero
University of Luxembourg
Luxembourg

ABSTRACT

This project studies whether aligned company-news event states improve multi-horizon excess-return forecasting beyond strong market baselines. I built a local, reproducible artifact pipeline for the top-50 liquid U.S. equities, with same-day-as-close news alignment, no-news states, purged expanding folds, an untouched lockbox, fold-local preprocessing, saved predictions, tuning ledgers, and conservative claim gates. The strongest current active model is an optimal model XGBoost configuration: 800 development trials produced a post-selection lockbox weighted error delta of -5.34×10^{-4} versus the prior XGBoost baseline. An optimal mixed model (frozen-base horizon mixer) improved validation error, but degraded versus optimal model XGBoost on lockbox, illustrating selection pressure. A historical portfolio simulation is reported only as a diagnostic. The main conclusion is methodological: in noisy financial forecasting, leakage control and non-claiming uncertainty analysis are as important as model complexity.

1 INTRODUCTION

Financial news may contain useful short-lived information, but return prediction is low signal-to-noise and highly vulnerable to leakage, multiple testing, and post-hoc trading narratives. The project therefore frames the task as end-of-day forecasting of forward excess returns for horizons $\{1, 5, 10, 20, 60\}$ days, not as price prediction or trading advice. The contribution is a reproducible deep-learning-ready benchmark: top-50 equities, Alpha Vantage news, Twelve Data OHLCV, explicit no-news rows, purged walk-forward development folds, a final lockbox, and artifact-backed model comparisons. The implemented roster spans linear and tree market baselines, FinBERT feature sidecars, TSMixer sequence diagnostics, optimal model XGBoost tuning, and horizon-level XGBoost/TSMixer mixing. AUC is intentionally not reported: the target is continuous excess return, so RMSE/MAE, zero-baseline R^2, rank IC, bootstrap support, and lockbox deltas are the relevant metrics.

2 RELATED WORK

The design follows market-efficiency and asset-pricing cautions from Fama, Fama-French, and Campbell-Lo-MacKinlay [4, 7, 8]: small apparent edges require strict out-of-sample controls. Textual-finance work shows why news can matter but also why representation matters [15, 17]. Modern NLP motivates contextual encoders and FinBERT-style finance adaptation [2, 6]. For tabular market features, gradient boosting remains a strong baseline [5, 12]. For sequence modeling, the relevant state of the art includes interpretable multi-horizon attention, patch-based transformers, inverted transformers, and foundation time-series models [1, 13, 14, 16]. Evaluation follows forecasting-metric and time-series validation guidance [3, 11]; multiple-testing caution follows Harvey et al. [10]. ROC-AUC is cited only to justify exclusion for this regression target [9].

3 EXPERIMENTS

Data and splits. The benchmark uses 50 liquid equities, five horizons, forward leave-one-out excess-return labels, and explicit event-state rows for no-news dates. The enhanced Alpha Vantage rebuild aligned 166,711 news rows while preserving provider-truncation limitations. Stage 4 defines nine expanding walk-forward development folds plus a held-out lockbox ending 2026-02-13; lockbox rows are never used for tuning.

Models and training. The main tuned model is the optimal model XGBoost event-market candidate, selected from 800 Optuna trials by a guarded validation objective over RMSE/MAE horizon deltas. The TSMixer branch exposed a readout bug: prior mean-time pooling was stale; the corrected readout is flattened time-feature state. Because a corrected full TSMixer search was estimated at 23.1 hours and was rejected, the optimal mixed model (frozen-base horizon mixer) tuned only a lightweight horizon mixer over frozen optimal model XGBoost and frozen optimal model TSMixer predictions; no TSMixer weights were retrained.

Results. Figure 1 shows the key pattern. Optimal model XGBoost improved against the prior XGBoost family on lockbox, but the optimal mixed model (frozen-base horizon mixer) is a cautionary counterexample: it improved validation and the frozen optimal model TSMixer diagnostic input, yet was worse than optimal model XGBoost on lockbox. Block-bootstrap evidence was supportive for the earlier frozen XGBoost candidate, but final claim gates remain conservative.

4 DISCUSSION

The strongest lesson is that financial deep learning must be evaluated as a system, not as a model leaderboard. The project found real engineering gains: leakage-aware splits, saved predictions, uncertainty diagnostics, tuning traces, and stronger market baselines. It also found failure modes: validation-to-lockbox degradation and portfolio-like diagnostics that can look persuasive while remaining non-causal and assumption-dependent. A practical bottleneck remains data compilation: API limits (25 calls/day) slow dataset refresh, so webscraping is advised to expand coverage. The transparent conclusion is therefore narrow: the optimal model XGBoost pipeline is the current strongest active evidence; the optimal mixed model (frozen-base horizon mixer) did not earn a positive lockbox claim; event-state value is not proven; and all outputs remain historical diagnostics, not live forecasts or trading signals.

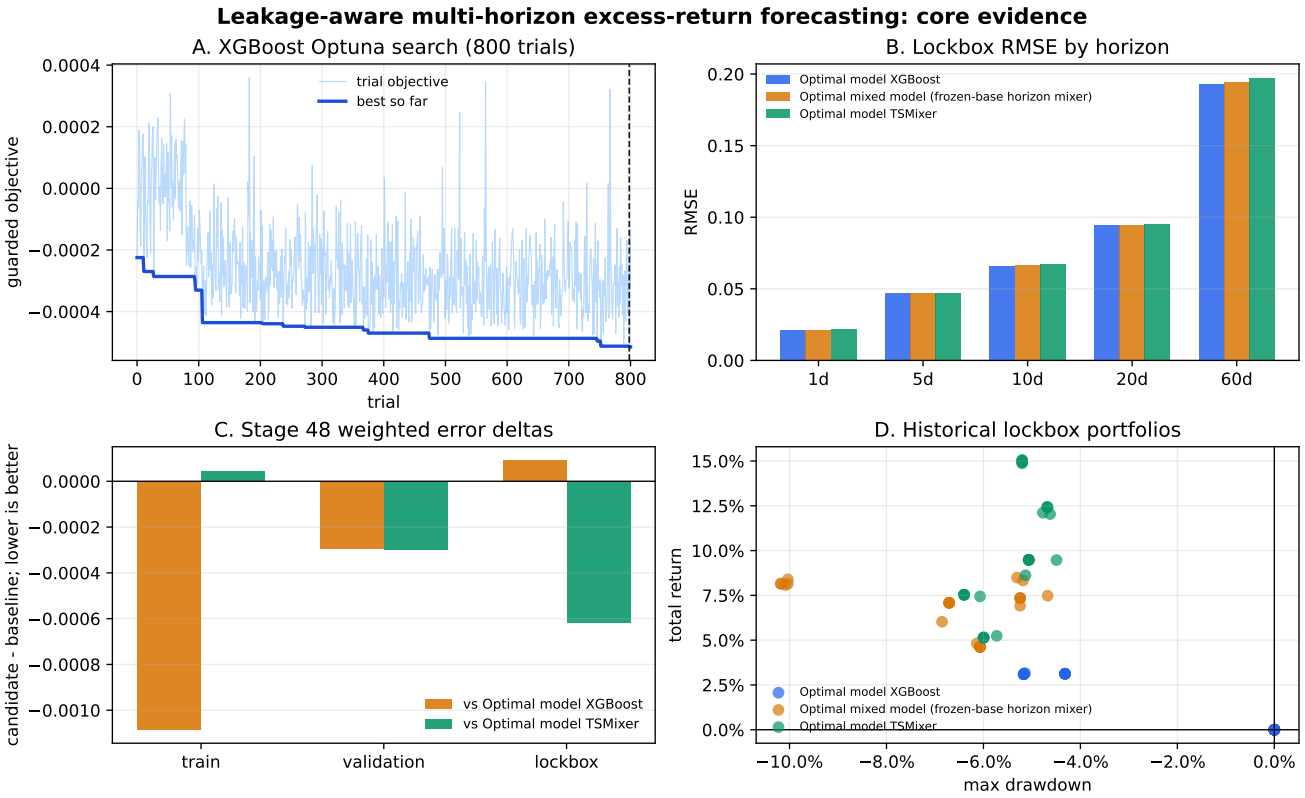


Figure 1: Core evidence. A: XGBoost tuning converged over 800 train/validation-only trials. B: lockbox RMSE by horizon. C: validation gains over optimal model XGBoost did not transfer to lockbox. D: the portfolio simulation is diagnostic only; high returns are not claim evidence.

REFERENCES

[1] Abdul Fatir Ansari, Lorenzo Stella, Caner Turkmen, Xiyuan Zhang, Pedro Mercado, Huibin Shen, Oleksandr Shchur, Syama Sundar Rangapuram, Sebastian Pineda-Arango, Shubham Kapoor, Jasper Zschiegner, Danielle C. Maddix, Michael W. Mahoney, Kari Torkkola, Andrew Gordon Wilson, Michael Bohlke-Schneider, and Yuyang Wang. 2024. Chronos: Learning the Language of Time Series. Transactions on Machine Learning Research. https://openreview.net/forum?id=gerNCVqqtR

[2] Dogu Araci. 2019. FinBERT: Financial Sentiment Analysis with Pre-trained Language Models. arXiv:1908.10063 [cs.CL] https://arxiv.org/abs/1908.10063

[3] Christoph Bergmeir, Rob J. Hyndman, and Bonsoo Koo. 2018. A Note on the Validity of Cross-validation for Evaluating Autoregressive Time Series Prediction. *Computational Statistics and Data Analysis* 120 (2018), 70–83. https://doi.org/10.1016/j.csda.2017.11.003

[4] John Y. Campbell, Andrew W. Lo, and A. Craig MacKinlay. 1997. *The Econometrics of Financial Markets*. Princeton University Press, Princeton, NJ, USA.

[5] Tianqi Chen and Carlos Guestrin. 2016. XGBoost: A Scalable Tree Boosting System. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*. ACM, New York, NY, USA, 785–794. https://doi.org/10.1145/2939672.2939785

[6] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. In *Proceedings of NAACL-HLT*. Association for Computational Linguistics, Minneapolis, Minnesota, 4171–4186. https://doi.org/10.18653/v1/N19-1423

[7] Eugene F. Fama. 1970. Efficient Capital Markets: A Review of Theory and Empirical Work. *The Journal of Finance* 25, 2 (1970), 383–417. https://doi.org/10.2307/2325486

[8] Eugene F. Fama and Kenneth R. French. 1993. Common Risk Factors in the Returns on Stocks and Bonds. *Journal of Financial Economics* 33, 1 (1993), 3–56. https://doi.org/10.1016/0304-405X(93)90023-5

[9] Tom Fawcett. 2006. An Introduction to ROC Analysis. *Pattern Recognition Letters* 27, 8 (2006), 861–874. https://doi.org/10.1016/j.patrec.2005.10.010

[10] Campbell R. Harvey, Yan Liu, and Heqing Zhu. 2016. … and the Cross-Section of Expected Returns. *The Review of Financial Studies* 29, 1 (2016), 5–68. https://doi.org/10.1093/rfs/hhv059

[11] Rob J. Hyndman and Anne B. Koehler. 2006. Another Look at Measures of Forecast Accuracy. *International Journal of Forecasting* 22, 4 (2006), 679–688. https://doi.org/10.1016/j.ijforecast.2006.03.001

[12] Guolin Ke, Qi Meng, Thomas Finley, Taifeng Wang, Wei Chen, Weidong Ma, Qiwei Ye, and Tie-Yan Liu. 2017. LightGBM: A Highly Efficient Gradient Boosting Decision Tree. In *Advances in Neural Information Processing Systems*. Curran Associates, Inc., Red Hook, NY, USA, 3146–3154.

[13] Bryan Lim, Sercan O. Arik, Nicolas Loeff, and Tomas Pfister. 2021. Temporal Fusion Transformers for Interpretable Multi-horizon Time Series Forecasting. *International Journal of Forecasting* 37, 4 (2021), 1748–1764. https://doi.org/10.1016/j.ijforecast.2021.03.012

[14] Yong Liu, Haixu Wu, Jianmin Wang, and Mingsheng Long. 2024. iTransformer: Inverted Transformers Are Effective for Time Series Forecasting. In *International Conference on Learning Representations*. OpenReview.net, Online, 1 pages. https://openreview.net/forum?id=JePfAI8fah

[15] Tim Loughran and Bill McDonald. 2011. When Is a Liability Not a Liability? Textual Analysis, Dictionaries, and 10-Ks. *The Journal of Finance* 66, 1 (2011), 35–65. https://doi.org/10.1111/j.1540-6261.2010.01625.x

[16] Yuqi Nie, Nam H. Nguyen, Phanwadee Sinthong, and Jayant Kalagnanam. 2023. A Time Series is Worth 64 Words: Long-term Forecasting with Transformers. In *International Conference on Learning Representations*. OpenReview.net, Online, 1 pages. https://openreview.net/forum?id=Jbdc0vTOcol

[17] Paul C. Tetlock. 2007. Giving Content to Investor Sentiment: The Role of Media in the Stock Market. *The Journal of Finance* 62, 3 (2007), 1139–1168. https://doi.org/10.1111/j.1540-6261.2007.01232.x

A ARTIFACT ROOTS AND REPLICATION COMMANDS

Primary evidence roots are the optimal model XGBoost tuning data/v1p0_full_results/v1p0__phase1_top50_updated_av__stage43__xgboost_tuning__20260528T100000Z__src_6dff1bd__run_40f1c9a0/, the frozen-base horizon mixer data/v1p0_full_results/v1p0__phase1_top50_updated_av__stage48__frozen_tsmixer_horizon_mixer__20260529T093301Z__src_6dff1bd__run_fc4c1756/, and the portfolio simulation diagnostic data/v1p0_full_results/v1p0__phase1_top50_updated_av__stage47__lockbox_portfolio_simulation__20260529T095157Z__src_6dff1bd__run_8317929b/. Validation command: `source ~/venvs/ai/bin/activate && python -m pytest tests`.

B SUPPLEMENTARY VISUALIZATIONS

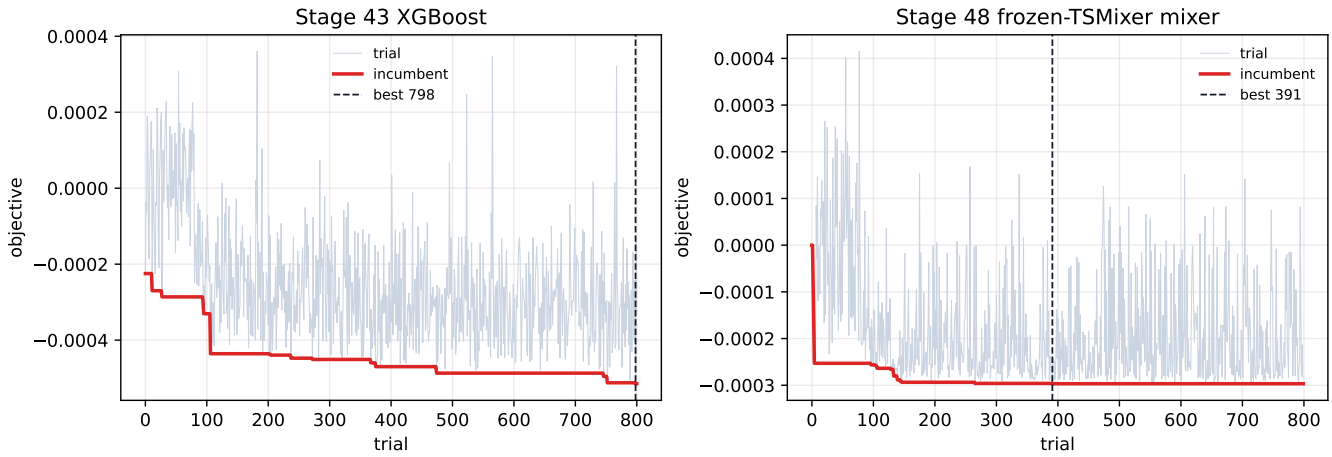


Figure 2: Optuna incumbent curves for optimal model XGBoost and the frozen-base horizon mixer.

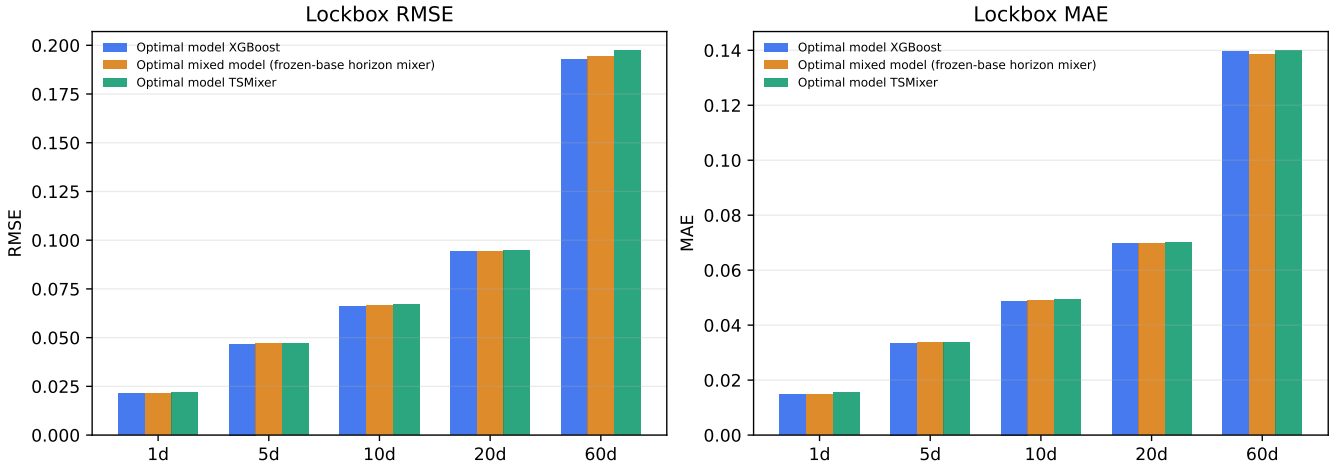


Figure 3: Lockbox RMSE and MAE by horizon for optimal model XGBoost, frozen optimal model TSMixer diagnostic input, and the frozen-base horizon mixer.

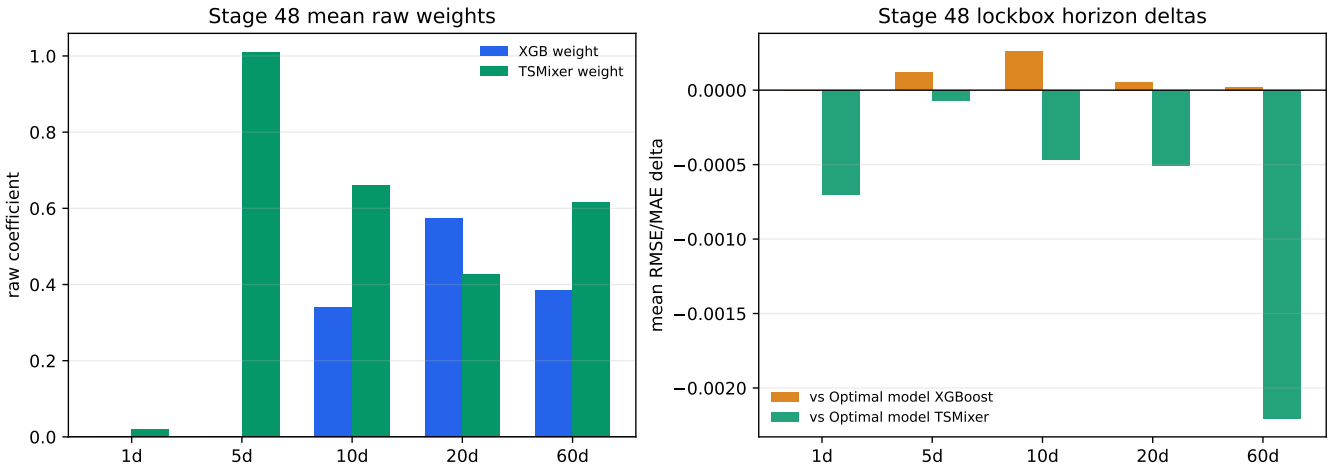


Figure 4: Horizon-specific raw mixer weights and lockbox horizon deltas.

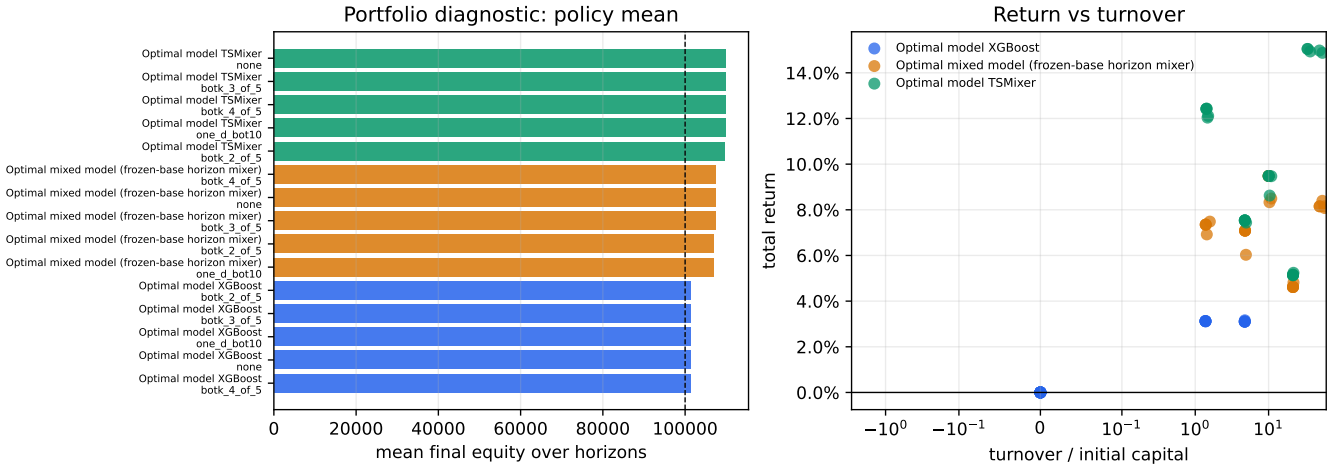


Figure 5: Historical portfolio diagnostics: final equity by policy and return versus turnover.

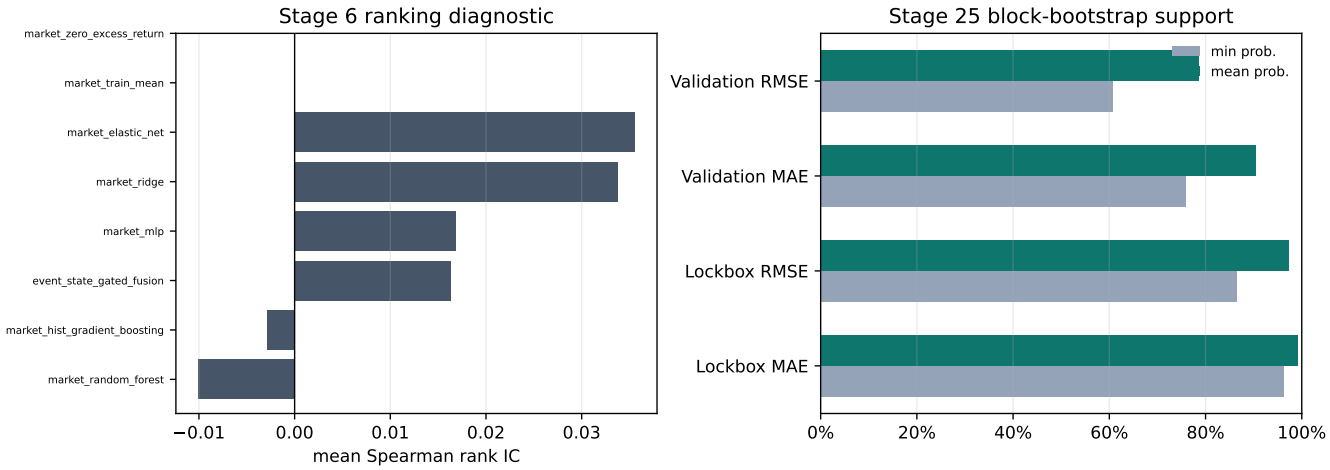


Figure 6: Ranking diagnostics and block-bootstrap support retained as secondary evidence.